Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models now answer hard scientific questions, yet they still hallucinate and follow broken lines of reasoning. Multi-agent debate offers one way to push back on this. When several models argue a question out before committing to an answer, the reasoning usually gets sharper and errors get caught.

The weakness is in how these debates settle. Most systems pick a winner by majority vote or by which agent sounds most confident. When a confident majority is simply wrong, it can pull a correct minority agent off its answer, and the group lands on the wrong conclusion. No current method checks whether an agent's claims actually hold up against outside scientific evidence before letting that agent sway the room.

We address this with a Trust-Calibrated Multi-Agent Scientific Deliberation framework that adjusts how much each agent counts as the debate unfolds.

The framework breaks every agent response into small factual claims. For each claim it retrieves relevant scientific literature and checks whether the evidence backs the claim or contradicts it. Every agent then carries a trust score that moves up or down as these checks accumulate over the rounds of debate. The final answer is built from trust-weighted aggregation rather than a raw count of votes.

Tying trust to real evidence means unsupported claims lose weight, so a correct but outnumbered agent is far less likely to be silenced.

We expect the approach to cut hallucination and keep reasoning tied to verifiable sources. It should also stop the debate from settling too early on a wrong answer, giving more reliable scientific results, and it does this without any model fine-tuning.

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Publication List

The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

Journal Articles

1. Leveraging Machine Learning in Balancing Inventories and Analyzing Market Trends: A Comprehensive Systematic Review (Artificial Intelligence Review, Under processing)

Conference Papers

1.

Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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Chapter 1

Introduction

This chapter introduces the project. It presents the background and problem, the motivation behind the work, the objectives, a short summary of the methodology, the expected outcome, and the structure of the rest of the report.

1.1 Project Overview

Large language models now handle many reasoning tasks, from scientific question answering to tutoring and decision support. A single model can still be confidently wrong. It may state an incorrect answer in fluent and assured language, which makes the mistake hard to catch. Multi-agent debate was proposed to reduce this problem. Several agents answer the same question, read each other's reasoning, and revise their positions over a few rounds before a final answer is selected [1]. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2], which makes the reasoning inside a debate easier to inspect. The real difficulty is the final step, where the separate answers are combined. Most debate systems settle disagreements by majority vote. Every agent gets equal weight, and the most common answer wins. This is simple, but it treats popularity as a stand-in for correctness. In scientific reasoning the better answer is the one backed by evidence, not the one repeated by more agents. The study examines a trust-calibrated multi-agent deliberation framework that adjusts each agent's influence during the debate based on how well its claims match retrieved external evidence. The final answer then depends on evidence-grounded trust rather than a raw headcount.

1.2 Motivation

Majority voting can fail in a way that is easy to miss. A debate may look like it reached agreement while it has quietly dropped a correct answer held by one agent. Recent studies of multi-agent debate report that agents often copy or switch answers under social pressure instead of reasoning on their own, and that the correct answer is present in the discussion but ignored in more than twenty percent of wrong-answer cases [3]. Work on model behaviour points to a related tendency, where models trained to be agreeable grow more sycophantic and less accurate [4]. When a confident majority pushes a correct minority agent to give up its answer, majority voting does more than pick the wrong result. It hides the failure behind an appearance of consensus. This matters most for scientific questions, where a wrong but confident consensus can mislead a user into trusting an answer that deserved a closer check. Existing methods approach this failure from different directions. iMAD decides when a debate is worth running [5], DebUnc uses self-reported confidence to adjust agent influence [6], and Mixture-of-Agents combines outputs through static aggregation [7]. These methods are useful, yet none ties an agent's influence to whether its claims hold up against external evidence while the debate is still active. That gap is what motivates the study.

1.3 Objectives

The main objective of this project is to reduce sycophantic consensus in multi-agent LLM debate by replacing plain majority voting with evidence-grounded trust calibration. This objective involves studying how majority-vote debate fails when a confident wrong majority pressures a correct minority agent into changing its answer, designing a bounded trust score that rises when an agent's claims are supported by external evidence and falls when they are contradicted, and applying that score during the debate before the final answer is fixed. The work also compares the proposed method with standard majority voting and other relevant debate and aggregation baselines, then evaluates the framework using answer accuracy together with consensus-collapse, minority-preservation, and evidence-calibration metrics.

1.4 Methodology

The study follows an experimental methodology. A baseline multi-agent debate system is built first, where several heterogeneous LLM agents answer the same scientific question, share their reasoning, and revise across rounds. The baseline uses majority voting so the usual failure mode can be observed and measured. Each agent's reasoning is then broken into smaller factual claims. These claims are checked against scientific evidence retrieved from external sources such as academic papers and reference databases. Any contradiction between a claim and the retrieved evidence is detected and scored [8], building on the idea that external tools can verify and correct model output [9]. The verdicts are turned into trust updates. Supported claims raise an agent's trust, contradicted claims lower it, and uncertain claims are handled conservatively. The trust score is bounded so that no agent gains unlimited influence or drops out of the debate completely. The final answer is chosen through trust-weighted aggregation rather than a simple count. The framework is evaluated on scientific reasoning datasets such as GPQA and MMLU-Pro. The evaluation checks whether the mechanism lowers consensus collapse, keeps correct minority answers, and holds overall accuracy without any model fine-tuning.

1.5 Project Outcome

The main outcome is a reproducible framework for trust-calibrated multi-agent scientific deliberation. It shows how retrieved evidence can update agent influence during a debate, and how the final answer can be decided by evidence-weighted trust instead of a majority vote. Alongside it, the study produces an evaluation protocol that detects sycophantic consensus, measures how often correct minority answers survive, and compares the method against established baselines. The study is expected to show whether dynamic trust calibration makes multi-agent debate more dependable on scientific questions. Even where the method does not remove every wrong answer, it should make the decision more transparent by recording which agents were supported or contradicted by the evidence.

1.6 Organization of the Report

The rest of this report is arranged as follows. Chapter 2 covers the background, the related work on multi-agent debate and sycophancy, and the gap this project addresses. Chapter 3 presents the requirement analysis, the detailed methodology, the system design, and the project plan. Chapter 4 describes the implementation setup, the testing and evaluation procedure, and the results with discussion. Chapter 5 discusses the relevant standards, the impact of the work, sustainability, and the complex engineering problem analysis. Chapter 6 concludes the report, notes its limitations, and gives recommendations for future work.

Chapter 2

Background

This chapter introduces the concepts needed to understand the proposed system and reviews research on multi-agent debate, sycophancy, and agent influence. It then explains the gap addressed by this project.

2.1 Preliminaries

Large language models (LLMs) generate answers one token at a time, but their outputs can still contain unsupported or incorrect reasoning. Fluent wording does not guarantee that an answer is correct. Chain-of-thought prompting asks a model to expose intermediate reasoning steps before giving a final answer, which makes the reasoning easier to inspect and gives later components more information to evaluate [2]. These steps are useful evidence about how an answer was formed, but they should not be treated as proof of correctness by themselves. Multi-agent debate extends this process by asking several agents to solve the same problem independently. The agents then receive the other answers and explanations, discuss their differences, and revise their positions over a fixed number of rounds. A final answer is selected after the discussion [1]. Several agents are useful because their independent attempts can expose different errors and allow one agent to question another agent's reasoning. The way those answers are combined still matters. A common method is majority voting, where each agent has the same weight and the most frequent answer is selected. If several agents repeat the same mistake, the vote can strengthen the mistake instead of correcting it. In this report, *sycophantic consensus* refers to an agent copying or abandoning a correct position because a wrong position has greater support. Confidence may be supplied by an agent or estimated from its token distribution, but neither signal directly verifies a claim. DebUnc shows this difference by comparing deployable uncertainty measures with a diagnostic Ground Truth measure that uses the correct answer [6]. The proposed framework therefore defines trust through support from retrieved scientific evidence. Zhang et al. (2025) provide a wider survey of reasoning- oriented LLMs [10].

2.2 Literature Review

The review covers studies that address debate construction, external verification, sycophancy, uncertainty, and debate evaluation. The main comparison is between methods that decide when to debate, methods that change agent interaction, and the proposed method that changes agent influence using retrieved evidence.

2.2.1 Similar Applications

The system is not a conventional web or mobile application. It is an experimental deliberation layer that could support a scientific question-answering tool, a research assistant, or another decision-support interface. Its closest similar systems are multi-agent LLM applications that generate several candidate answers and combine them before returning a result. The difference is that this project also records evidence checks and the changing trust of each agent.

2.2.2 Related Research

Wei et al. (2022) showed that chain-of-thought prompting can improve reasoning in large language models [2]. Du et al. (2024) extended this idea through multi-agent debate, where agents answer a question, examine peer reasoning, and revise their positions [1]. Gou et al. (2024) proposed CRITIC, which uses external tools to check and improve model responses [9]. The method supports external verification, but it does not use several models debating the same question. Wang et al. (2025) introduced Mixture-of-Agents, a layered method that combines outputs from several proposer models [7]. It shows the value of multiple LLMs, but it uses static aggregation without an explicit evidence-based trust value. Fan, Yoon, and Ji (2026) proposed iMAD to reduce the cost of running debate on every question [5]. Their method extracts 41 features from a structured self-critique and uses a classifier to decide whether debate should start. It reports token savings of up to 92 percent compared with GroupDebate and an accuracy improvement of up to 13.5 percentage points over a single-agent baseline. However, it does not change agent influence after debate begins. Pitre, Ramakrishnan, and Wang (2025) studied sycophancy inside multi-agent debate [3]. Their work found that the correct answer can be present but ignored in more than 20 percent of wrong-answer cases. CONSENSAGENT detects stalling and answer copying, then rewrites the task prompt before another debate stage. It reduces measured sycophancy, but its final aggregation still uses self-reported confidence and consistency. Yoffe, Amayuelas, and Wang (2025) proposed DebUnc, which shares agent uncertainty through prompts or attention scaling during debate [6]. Its deployable uncertainty measures give small gains, while a diagnostic Ground Truth signal gives a much larger gain. This result shows that the trust signal itself is a major limitation. Kushwaha and Madhavan (2026) described an eight-agent MAS-RAG architecture that detects contradictions among retrieved documents and ranks evidence by credibility [8].

Its source credibility is fixed by a human curator and is not updated during deliberation. Cau et al. (2025) examined opinion convergence in LLM interactions and found that selective persuasion can influence convergence [11]. Yeow et al. (2025) compared several decision architectures and reported that feedback between agents can allow early errors to spread [12]. Wang et al. (2024) found that a single LLM with strong in-context examples can sometimes match multi-agent discussion systems, and they identified judge errors and peer conformity as weaknesses [13]. Choi, Zhu, and Li (2025) separated debate communication from voting and argued that majority voting accounts for much of the usual improvement [14]. Cui et al. (2026) proposed FREE-MAD, which avoids final consensus and uses anti-conformity prompts with trajectory scoring [15]. Zhou et al. (2025) proposed ReSo, a reward-driven system that selects agents through dynamic task graphs [16]. These works study coordination and agent selection, but they do not verify scientific claims before changing agent influence. Ibrahim et al. (2026) reported that tuning models to sound warm and agreeable can reduce factual accuracy and increase sycophancy [4]. Pitre et al. (2026) proposed process-level diagnostics for engagement, responsiveness, influence asymmetry, balance, stability, and agent utility in debates where ground truth may be unavailable [17]. Their work shows that final agreement alone is not enough to judge a debate. Overall, the literature provides methods for debate, external checking, uncertainty sharing, prompt correction, and process evaluation. The remaining gap is a system that continuously updates an agent's influence during scientific deliberation according to claim-level support from retrieved evidence. This project addresses that gap with a bounded trust score.

2.3 Gap Analysis

The reviewed papers leave three needs in multi-agent debate. A system must decide whether the extra cost of debate is justified, detect when a discussion is moving toward unsupported agreement, and decide which agent should have more influence when the agents disagree. iMAD mainly addresses the first need. CONSENSAGENT addresses the second through prompt optimization. DebUnc addresses the third with an in-debate weighting mechanism, but its practical signals are internal uncertainty estimates. The remaining gap is an in-session trust update based on independently checkable evidence. The proposed framework addresses this gap by decomposing an agent's response into smaller claims, retrieving relevant scientific passages, and classifying each claim as supported, contradicted, or unverifiable. The verdicts update a bounded trust score after each debate round. An agent can therefore be in the numerical majority and still lose influence if the claims supporting its position are contradicted. A correct minority agent can retain influence when its claims are better supported. No evidence is recorded as an unresolved case, while direct contradiction is handled separately. The evaluation therefore uses consensus-collapse rate, minority-preservation rate, and evidence-calibration rate alongside ordinary answer accuracy. Retrieval sparsity, imperfect claim decomposition, and correlated errors between agents remain limitations of the proposed approach.

2.4 Summary

This chapter introduced LLM reasoning, chain-of-thought prompting, multi-agent debate, and sycophantic consensus. The literature review covered debate efficiency, prompt correction, uncertainty weighting, external verification, process evaluation, and agent coordination. The reviewed methods do not continuously update agent influence from claim-level scientific evidence. The next chapter uses this gap to define the system requirements, detailed methodology, and project plan.

Chapter 3

Project Design

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

3.1 Requirement Analysis

3.1.1 Functional and Nonfunctional Requirements

3.1.2 Context Diagram

3.1.3 Data Flow Diagram Level 1

3.1.4 UI Design

3.2 Detailed Methodology and Design

You have to mention alternate solutions that you have considered. Why you have selected the specific solution, etc.

3.3 Project Plan

3.4 Task Allocation

3.5 Summary

Chapter 4

Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

4.1 Environment Setup

4.2 Testing and Evaluation

4.3 Results and Discussion

4.4 Summary

Chapter 5

Standards and Design Constraints

[Must be present in FYDP-1 Report and also in Final Report]

Every chapter should start with 1-2 sentences on the outline of the chapter.

5.1 Compliance with the Standards

Only mention the standards that are related to your project. This list is not complete. For each of the standards discuss the alternates with pros and cons and rationale of selection.

5.1.1 Software Standards

5.1.2 Hardware Standards

5.1.3 Communication Standards

5.2 Design Constraints

Only mention the constraints that are related to the design of your project. This list is not complete.

5.2.1 Economic Constraint

5.2.2 Environmental Constraint

5.2.3 Ethical Constraint

5.2.4 Health and Safety Constraint

5.2.5 Social Constraint

5.2.6 Political Constraint

5.2.7 Sustainability

5.3 Cost Analysis

Provide a cost analysis in terms of budget required and revenue model. In case of budget, you must show an alternate budget and rationales.

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.1: Mapping with complex problem solving.

P1 Dept of Knowledge	P2 Range of Conflicting Requirements	P3 Depth of Analysis	P4 Familiarity of Issues	P5 Extent of Applicable Codes	P6 Extent of Stakeholder Involvement	P7 Interdependence
√	√					

5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.2).

5.5 Summary

Table 5.2: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of Interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
√	√			

Chapter 6

Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

6.1 Summary

6.2 Limitation

6.3 Future Work

References

[1] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 11733–11763, 2024.

[2] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA, 2022. Curran Associates Inc.

[3] Priya Pitre, Naren Ramakrishnan, and Xuan Wang. CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 22112–22133, Vienna, Austria, July 2025. Association for Computational Linguistics.

[4] L. Ibrahim, F. S. Hafner, and L. Rocher. Training language models to be warm can reduce accuracy and increase sycophancy. *Nature*, 652:1159–1165, April 2026.

[5] Wei Fan, JinYi Yoon, and Bo Ji. iMAD: Intelligent multi-agent debate for efficient and accurate LLM inference. In *Proceedings of the Fortieth AAAI Conference on Artificial Intelligence*, pages 29403–29411, 2026.

[6] Luke Yoffe, Alfonso Amayuelas, and William Yang Wang. DebUnc: Improving large language model agent communication with uncertainty metrics. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 23299–23315, Suzhou, China, November 2025. Association for Computational Linguistics.

[7] Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Y Zou. Mixture-of-agents enhances large language model capabilities. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 33944–33963, 2025.

[8] Pranjal Kushwaha and S. Akshat Madhavan. Agent-based contradiction detection and resolution for multi-source retrieval augmented generation systems. *International*

Journal of Scientific Research in Engineering and Management (IJSREM), 10(7), July 2026.

[9] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *The Twelfth International Conference on Learning Representations*, 2024.

[10] D. Zhang, Z.-Z. Li, M.-L. Zhang, J. Zhang, Z. Liu, Y. Yao, H. Xu, J. Zheng, X. Chen, Y. Zhang, F. Yin, J. Dong, Z. Guo, L. Song, and C.-L. Liu. From system 1 to system 2: a survey of reasoning large language models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–20, 2025.

[11] E. Cau, V. Pansanella, D. Pedreschi, et al. Selective agreement, not sycophancy: investigating opinion dynamics in LLM interactions. *EPJ Data Science*, 14:59, August 2025.

[12] Xian Yeow, Shunichi Lee, Lasitha Akatsuka, Vidyaratne Aman, A Hareesh Kumar, Chetan Farahat, and A Gupta. Reliable decision-making for multi-agent LLM systems. *arXiv preprint arXiv:2406.04092*, 2025.

[13] Qineng Wang, Zihao Wang, Ying Su, Hanghang Tong, and Yangqiu Song. Rethinking the bounds of LLM reasoning: Are multi-agent discussions the key? In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6106–6131, Bangkok, Thailand, August 2024. Association for Computational Linguistics.

[14] Hyeong Kyu Choi, Jerry Zhu, and Sharon Li. Debate or vote: Which yields better decisions in multi-agent large language models? In *Advances in Neural Information Processing Systems*, 2025.

[15] Yu Cui, Hang Fu, Haibin Zhang, Licheng Wang, and Cong Zuo. FREE-MAD: Consensus-free multi-agent debate. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026.

[16] Heng Zhou, Hejia Geng, Xiangyuan Xue, Li Kang, Yiran Qin, Zhiyong Wang, Zhenfei Yin, and Lei Bai. Reso: A reward-driven self-organizing llm-based multi-agent system for reasoning tasks. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 15979–15998. Association for Computational Linguistics, 2025.

[17] Priya Pitre, Gaurav Srivastava, Lu Zhang, Le Wang, Naren Ramakrishnan, and Xuan Wang. A diagnostic study of multi-agent llms for real-world debates. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *PMLR*, Seoul, South Korea, 2026.